

Course Code: B23BS2105					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. I Semester MODEL QUESTION PAPER					
PROBABILITY THEORY AND STOCHASTIC PROCESS					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define joint probability and conditional probability.	1	1	2
	b).	Find the chance of getting a total of exactly 9 in throwing 2 dice.	1	3	2
	c).	Define characteristic function of a random variable?	2	1	2
	d).	Find the expected value of the random variable X which is uniformly distributed between 2 and 8.	2	3	2
	e).	write the properties of joint density function?	3	1	2
	f).	Define central limit theorem.	3	1	2
	g).	State Wiener-Khinchine Theorem.	4	1	2
	h).	Which one is valid PSD (a). $3 + j\omega^2$ (b). $18\delta(\omega)$	4	3	2
	i).	Define a time invariant system.	5	1	2
	j).	Define white noise.	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the terms probability & axioms of probability.	1	2	5
	b).	In a bolt factory, machines A, B, C manufacture 30%,30%,40%, of the total output respectively. From their outputs 4,5,3 percent are defective items produced by each machine respectively. Now an item selected which is to be defective. What is the probability that it was manufactured by machines B.	1	3	5
		OR			
3.	a).	Explain the following density functions. i). Exponential and ii). Rayleigh	1	2	5
	b).	The pdf of a continuous random variable is given by $f_X(x) = \begin{cases} C(x-1) & 1 \leq x \leq 4 \\ 0 & \text{elsewhere} \end{cases}$ (i) Find the value of constant 'C' (ii) find $P\{2 \leq X \leq 3\}$	1	3	5
		UNIT-2			
4.	a).	State and prove any four properties of expectation.	2	3	5

	b).	What is the probability density function of uniform random variable. X whose mean $E[X] = 7$ and $\text{var}[x] = 3$.	2	3	5
		OR			
5.	a).	State and prove chebyshev's inequality	2	3	5
	b).	Consider that a pdf of a random variable X is $f_x(x) = 1/K \quad -2 \leq x \leq 3$ and another random variable $Y = 2X$ $= 0$ otherwise Then find value of K, $E[X]$, $E[Y]$ and $E[XY]$.	2	3	5
		UNIT-3			
6.	a).	State and prove any four properties of joint probability distribution function	3	3	5
	b).	The joint probability density function of two random variables X and Y is given by $f_{XY}(x,y) = k e^{-(ax+by)}$ if $x \geq 0, y \geq 0$, $= 0$ otherwise Find the value of k. Also prove that X and Y are independent.	3	3	5
		OR			
7.	a).	Define joint characteristic function and list their properties.	3	2	5
	b).	Obtain the relation between joint moments and joint central moments.	3	3	5
		UNIT-4			
8.	a).	A random process is given as $X(t) = At$, where A is uniformly distributed random variable on (0,2).find whether X(t) is wide sense stationary or not.	4	3	5
	b).	Explain the following. i). Stationary process and ii). Ergodicity.	4	2	5
		OR			
9.	a).	Explain power spectral density of a random process and list their properties.	4	2	5
	b).	The auto correlation fnction of a random process is $R_{xx}(\tau) = (A^2/2) \cos \omega \tau$. Determine the PSD.	4	3	5
		UNIT-5			
10.		Explain in detail with mathematical expressions (i) Mean value of the response of a system (ii) The auto correlation fnction of response of a system.	5	3	10
		OR			
11.		Find the input auto correlation fnction ,output auto correlation fnction and output spectral density of RC low pass filter when the filter is subjected to a white noise of spectral density $N_0/2$.	5	3	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23HS2101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. I Semester MODEL QUESTION PAPER					
UNIVERSAL HUMAN VALUES-2: UNDERSTANDING HARMONY AND ETHICAL HUMAN CONDUCT					
(Common to all programmes of Engineering)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What are the basic guidelines for value education?	1	2	2
	b).	What is MBTI personality test?	1	2	2
	c).	How can we differentiate between the needs of the Self and the needs of the Body?	2	2	2
	d).	What are the characteristics and activities of the Self (I)?	2	2	2
	e).	How is 'respect' defined in the context of human interaction?	3	2	2
	f).	How is society described in relation to the family?	3	2	2
	g).	How are the four orders of nature interconnected?	4	2	2
	h).	How does the idea of self-regulation in nature contribute to its harmony?	4	2	2
	i).	Define definitiveness of (ethical) human conduct.	5	2	2
	j).	Explain how humanistic education can influence professional ethics.	5	2	2
5 x 10 = 50 Marks					
		UNIT - I	CO	KL	M
2.	a).	Discuss natural acceptance	1	2	5
	b).	Differentiate prosperity and deprivation	1	2	5
		OR			
3.	a).	Deliberate the right understanding in perspective to self-exploration.	1	2	5
	b).	What are the key functions of the MBTI? Explain.	1	2	5
		UNIT - II			
4.	a).	Illustrate coexistence of "I" and "Body ".	1	2	5
	b).	Distinguishing between the Needs of the Self and the Body	1	2	5
		OR			
5.	a).	Discuss Characteristic activities of Harmony with "I".	1	2	5
	b).	Explain Sanyam and Health.	1	2	5

		UNIT - III			
6.	a).	Write a note on human-human relationship as regarding harmony.	2	2	5
	b).	Differentiate intention and competence.	2	2	5
		OR			
7.	a).	Discuss salient values in relationship.	3	2	5
	b).	Illustrate universal Harmonious Society - an Undivided society.	3	2	5
		UNIT - IV			
8.		Discuss orders of life in nature and its significance self regulation of individual	4	2	10
		OR			
9.		Illustrate existence of human being as coexistence with universe in perspective of space	4	2	10
		UNIT - V			
10.		Discuss importance of professional competence for augmenting universal human order.	5	2	10
		OR			
11.	a).	Case study of typical holistic technologies.	5	2	5
	b).	Role of engineer in promoting harmony in society	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

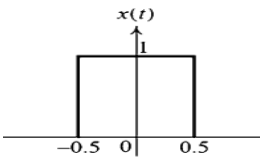
M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Estd. 1980

AUTONOMOUS

Course Code: B23EC2101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. I Semester MODEL QUESTION PAPER					
SIGNALS AND SYSTEMS					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
First Question is compulsory					
Answer all the questions. All questions carry equal marks					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain how a power signal is different from an energy signal with an example	1	2	2
	b).	What are the properties to be satisfied by the LTI system?	1	2	2
	c).	What are Dirichlet's conditions?	2	2	2
	d).	Give the relation between Exponential and Trigonometric Fourier series coefficients.	2	2	2
	e).	State the differentiation in time domain property of the continuous-time Fourier Transform	3	2	2
	f).	Let $x(t)=j/\pi t$, Evaluate its Fourier Transform	3	3	2
	g).	What is the relation between Laplace transform and Fourier transform?	4	2	2
	h).	An LTI system $h(t)$ is stable , Discuss ROC of system function $H(s)$.	4	2	2
	i).	State the Nyquist criterion for sampling continuous-time signals.	5	2	2
	j).	What is the time shifting property of Z transform?	5	2	2
Estd. 1980 AUTONOMOUS					
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Determine the energy and power value of the following signals. i) $x(t) = 5 \cos (50t + 3)$, ii) $x(t) = 2e^{- t }$	1	3	5
	b).	Explain all classification of signals with examples for each category.	1	3	5
		OR			
3.	a).	Determine whether the following systems are time invariant or not. i) $y(t) = t^2 x(t - 1)$ ii) $y(n) = x(2n-2)$	1	3	5
	b).	Consider $x(t)=u(t+1)-u(t)$, plot $y_1(t)=x(t-4)$ and $y_2(t)=x(-t-4)$	1	3	5
		UNIT-2			
4.	a).	Evaluate trigonometric Fourier series representation of the following signal with fundamental period $T=2$ sec. $x(t) = t + 1, for -1 < t < 0$ & $-t + 1, for 0 < t < 1$	2	3	5

	b).	Discuss the Gibbs phenomenon effect in Fourier series representation?	2	2	5
		OR			
5.	a).	Determine exponential F.S coefficients C_k of given $x(t)$, $x(t) = 2 + \cos\left(\frac{2\pi}{3}t\right) + \cos\left(\frac{5\pi}{3}t\right)$	2	3	5
	b).	State and prove time shifting property of the CT Fourier Series?	2	3	5
		UNIT-3			
6.	a).	State and prove the differentiation in the frequency domain property of CTFT.	3	3	5
	b).	Determine Fourier transform of $x(t) = e^{4t} \cos 4t$	3	3	5
		OR			
7.	a).	Find the Fourier transform of the following signal shown in figure below. 	3	3	5
	b).	If Fourier transform of $x(t)$ is $X(j\omega)$, then calculate the Fourier transform of $x_1(t) = x(3t - 6)$ by using FT properties	3	3	5
		UNIT-4			
8.	a).	State and prove the final value theorem of Laplace transform.	4	2	5
	b).	Evaluate the initial and final values for $X(s) = \frac{2s+5}{s(s+3)(s+4)}$	4	2	5
		OR			
9.	a).	Explain ROC properties of Laplace transforms.	4	2	5
	b).	Determine Laplace transform of $x(t) = te^{-2 t }$	4	3	5
		UNIT-5			
10		State and prove sampling theorem for low pass signals	5	2	10
		OR			
11.	a).	A causal LTI system is described by the following difference equation $y(n) - y(n-1) - y(n-2) = x(n-1)$. Verify whether the system is stable or not.	5	3	5
	b).	Evaluate inverse Z-Transform of $X(z) = \frac{z^2+2z+3}{(z-1)(z-3)(z-4)}$, for $ z > 4$ and $ z < 1$	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC2102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech I Semester(R23)MODELQUESTION PAPER					
ELECTRONIC DEVICES AND CIRCUITS					
For ECE					
Time:3Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Mobility and Conductivity.	1	1	2
	b).	Differentiate intrinsic and extrinsic semi conductors.	1	2	2
	c).	Write the advantages of CE configuration.	2	2	2
	d).	Draw output characteristics of CB transistor and indicate the regions.	2	2	2
	e).	Discuss thermal runaway.	3	2	2
	f).	Define stability factors S , S^I , S^{II}	3	1	2
	g).	Draw the h-parameter equivalent circuit of the CE amplifier.	4	2	2
	h).	Why low frequency gain of the CE amplifier decreases as frequency decreases?	4	2	2
	i).	Compare JFET with MOSFET.	5	2	2
	j).	Define Trans conductance.	5	1	2
Estd. 1980 AUTONOMOUS					
5 x 10 = 50 Marks					
		UNIT-1			
2.		Distinguish between intrinsic and extrinsic semiconductors.	1	4	10
		OR			
3.		Explain the operation and characteristics of p-n Diode in forward bias condition.	1	2	10
		UNIT-2			
4.		Distinguish between Zener breakdown and Avalanche breakdown.	2	4	10
		OR			
5.		With a neat diagram, explain the operation of a full wave rectifier and obtain expressions for ripple factor and efficiency.	2	3	10
		UNIT-3			
6.		Compare CE, CC and CB configurations.	3	2	10
		OR			

7.		A transistor with $\beta = 100$ is to be used in Common Emitter Configuration with collector to base bias. $R_E = 1 \text{ K}\Omega$, $V_{CC} = 10\text{V}$. Assume $V_{BE} = 0.7\text{V}$. Choose R_B so that the quiescent collector to emitter voltage is 4V. Find Stability Factor S.	3	3	10
		UNIT-4			
8.		Analyze CB configuration using exact model.	4	4	10
		OR			
9.		Analyze CE amplifier using approximate h-parameter model.	4	4	10
		UNIT-5			
10.		Explain the operation of JFET with the help of Drain and Transfer characteristics.	5	3	10
		OR			
11.		Explain the operation of Enhancement and Depletion modes of MOSFET in detail.	5	3	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks



Course Code: B23EC2103					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. I Semester MODEL QUESTION PAPER					
SWITCHING THEORY& LOGIC DESIGN					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
First Question is compulsory					
Answer all the questions. All questions carry equal marks					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	List the classification of binary codes.	1	1	2
	b).	State De Morgan's laws.	2	1	2
	c).	Define minterm and maxterms.	2	1	2
	d).	Define combinational circuit?	3	1	2
	e).	Draw 4x1 Multiplexer using 2x1 Multiplexers	3	3	2
	f).	Distinguish between PROM, PLA, PAL.	3	2	2
	g).	Distinguish between transition table and excitation table.	4	2	2
	h).	What are the applications of shift registers?	4	1	2
	i).	Differentiate synchronous counters and Asynchronous counters,	5	2	2
	j).	What is the significance of State assignment?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	convert the following decimal numbers to the indicated base: i) 7562 to Octal ii) 1938 to hexadecimal	1	3	5
	b).	Convert the following expressions into sum of products and product of sum. (i). $(AB+C)(B+C'D)$	2	3	5
		OR			
3.	a)	Explain how you can detect a single error occurred during transmission in a 7 bit Hamming code word with an example.	1	3	5
	b)	For each of the following Boolean expressions, give the truth table a. $(a+(b+c))(c+(a+b))$ b. $wy+wxz$	2	3	5
		UNIT-2			
4.		Minimize the following function using Karnaugh map method and draw	2	3	5

		simplified logic diagram. $F = m(1,3,7,11,15) + d(0,2,5)$			
	b)	Design a circuit which converts Binary to Gray code.	3	4	5
		OR			
5.	a)	Minimize the following function using Karnaugh map method and draw Nor to Nor logic diagram. $F = \Pi(0,1,3,4,5,7,9,11,12,13,14,15)$	2	3	5
	b)	Design a logic diagram of a addition/subtraction. Use a control variable W and a circuit that functions as a full adder when $W=0$, as a full subtractor when $W=1$.	3	4	5
		UNIT-3			
6		Implement the given functions by using a 3 to 8 Decoder and logic Gates. (i) $f_1(A,B,C) = \Sigma m(0,1,5,7)$ (ii) $f_2(A,B,C) = \Sigma m(1,3,5,7)$	3	3	10
		OR			
7.	a).	Implement the following function with multiplexer $F(A,B,C,D) = (0,1,3,4,8,9,15)$	3	3	5
	b).	Design and implement Full adder with PLA.	3	3	5
		UNIT-4			
8.	a).	Convert D-Flip Flop into JK-Flip Flop.	4	3	5
	b).	What is a race around problem in a JK flip flop, avoid it by using Master Slave JK-Flip Flop.	4	2	5
		OR			
9.	a).	Draw and explain 4-bit universal shift register.	4	2	5
	b).	Design 3 bit right shift register and draw its output waveforms.	4	3	5
		UNIT-5			
10.		Design a 3 bit ripple counter using T-flip-flop and explain the operation.	5	4	10
		OR			
11.	a).	Explain the following related to sequential circuits with suitable examples. (a) State diagram. (b) State table. (c) State assignment.	5	2	6
	b).	Differentiate between moore and mealy circuits.	5	2	4

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

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Course Code: B23HS2201					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. II Semester MODEL QUESTION PAPER					
MANAGERIAL ECONOMICS AND FINANCIAL ANALYSIS					
(Common to AIDS, CSE, CIC, CSG, CSIT, CE, ECE, EEE, ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a)	Define Managerial Economics.	1	1	2
	b)	State the Importance of Demand forecasting.	1	1	2
	c)	Write about Fixed cost and Variable cost.	2	1	2
	d)	List out the Applications of Break-even analysis.	2	1	2
	e)	Define Double Entry System of Accounting.	3	1	2
	f)	List the items under Current assets and Current liabilities.	4	1	2
	g)	Name the types of Imperfect Competition.	5	1	2
	h)	Identify the methods of Internet Pricing.	5	1	2
	i)	Show the components of working capital cycle.	6	1	2
	j)	Write the importance of Depreciation.	6	1	2
5 x 10 =50Marks					
		UNIT-1	CO	KL	M
2.	a)	Compare the differences between Micro and Macro Economics.	1	2	5
	b)	Explain the Scope of Managerial Economics.	1	2	5
		OR			
3.	a)	Explain the determinants of Demand.	1	2	5
	b)	Describe the types of Elasticity of Demand.	1	2	5
		UNIT-2			
4.	a)	Illustrate the Elements of costs with suitable examples.	2	2	5
	b)	Define Cost. Explain the types of Costs.	2	2	5
		OR			
5.	a)	Interpret the determination of Break-even point with graphical representation.	2	2	5
	b)	Identify the Assumptions and Limitations of Break-even analysis.	2	2	5
		UNIT-3			
6.		Write the importance of Accounting and explain the types of accounts	3	2	10

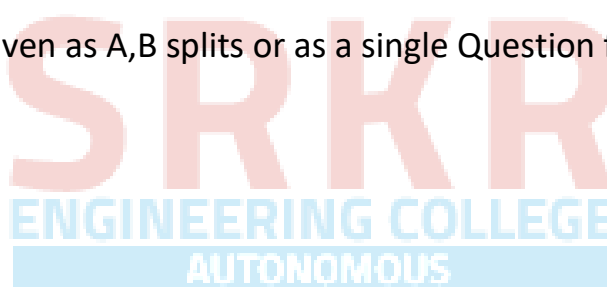
		with rules governing each account.			
		OR			
7.		Illustrate the proforma for Trading and Profit and loss account and Balance sheet including items in each account.	4	2	10
		UNIT-4			
8.	a)	Outline the salient features of Perfect competition.	5	2	5
	b)	Discuss the features of Oligopoly.	5	2	5
		OR			
9.	a)	Explain different methods of Cost Based Pricing.	5	2	5
	b)	Describe the Competition Based pricing methods.	5	2	5
		UNIT-5			
10.	a)	Discuss the factors influencing Working capital.	5	2	5
	b)	Explain the Sources of Raising finance in long term.	5	2	5
		OR			
11.	a)	Define Depreciation. Explain the causes of Depreciation in detail.	6	2	5
	b)	Explain the methods of Depreciation.	6	2	5

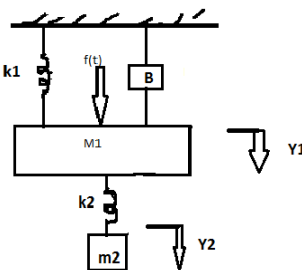
CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

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Course Code: B23EC2201					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. II Semester MODEL QUESTION PAPER					
LINEAR CONTROL SYSTEMS					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain how feedback effects overall gain of the system	1	2	2
	b).	State and explain Mason's gain formula.	1	2	2
	c).	What is the difference between type and order of a system?	2	1	2
	d).	Explain the effects of Integral control action?	2	2	2
	e).	What are the limitations of Routh's stability?	3	1	2
	f).	Compare between absolute stability, conditional stability and relative stability.	3	2	2
	g).	What are frequency domain specifications?	4	1	2
	h).	Define gain and phase margins.	4	1	2
	i).	What are the advantages of state variable techniques?	5	1	2
	j).	What are the properties of state transition matrix?	5	1	2
5 x 10 =50Marks					
		UNIT-1			
2.	a).	Compare open-loop and closed loop control systems with examples.	1	3	5
	b).	For the given mechanical system, obtain the transfer function, $Y_1(s)/F(s)$ and draw its electrical analog based on f-i analogy. 	1	4	5
		OR			
3.	a).	Find the transfer function C/R for the following system by using block diagram reduction technique	1	3	5

	b).	Obtain the transfer function y_6/y_1 for the given SFG using Mason's gain rule.	1	3	5
		UNIT-2			
4.	a).	Distinguish between type and order of a system. What are the various error constants and how they are related to the type of the system?	2	3	5
	b).	Derive the expression for unit-step response of a standard second-order system which is under-damped. Also, describe the transient response specifications.	2	4	5
		OR			
5.	a).	Illustrate the standard input signals used to test control systems. Which one is used mostly and why?	2	3	5
	b).	A unity feedback system is characterized by an open-loop transfer function $G(s) = K/S(S+10)$. Determine the value of gain K such that the system has a damping ratio of 0.5. With this value of K, find the settling time, percent overshoot and steady state error for a unit-ramp input.	2	4	5
		UNIT-3			
6.	a).	Illustrate why all the poles of a closed loop system must lie in the left-half of the s-plane for the system to be stable.	3	3	5
	b).	Using R-H criterion, calculate the range of K for the closed loop system to be stable. The open loop transfer function of the system is $G(S)H(S) = \frac{K}{S(S+1)(S+2)}$	3	4	5
		OR			
7.	a).	Explain how 'Relative stability' of a system can be assessed using RH criterion?	3	3	5
	b).	Obtain the Root-locus for the system with	3	4	5

		$G(S)H(S) = \frac{K}{S(S+4)(S^2+4S+8)}$ Determine the value of K makes the closed loop system marginally stable?			
		UNIT-4			
8.	a).	List the frequency domain specifications? A second order system step response shows 25% overshoot. Determine its resonant peak in frequency response?	4	3	5
	b).	Obtain the Bode plots for the system having OL transfer function $G(S)H(S) = \frac{2500}{S(S+5)(S+50)}$ Determine the Gain Margin and Phase Margin.	4	3	5
		OR			
9.	a).	The open loop transfer function of a unity feedback system is given by $G(S) = \frac{1}{S(1+S)(1+2S)}$ Draw the polar plot.	4	4	5
	b).	Draw Nyquist diagram and determine the stability of a closed loop control system with open-loop transfer function $G(S)H(S) = \frac{3}{S(S+1)^2}$	4	3	5
		UNIT-5			
10.	a).	Obtain the SS model for the following transfer function. $G(s) = \frac{2S+1}{S^2+7S+9}$	5	4	5
	b).	Obtain the transfer function model for the following system $\dot{x} = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} x + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u$ $y = [1.5 \quad 0.625]x$	5	3	5
		OR			
11.	a).	Find the state transition matrix for the given system. $\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u$	5	3	5
	b).	Test the controllability and observability for the following SS model $\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u(t) \quad y = [3 \quad 4 \quad 1]x$	5	4	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC2202					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B. Tech. II Semester MODEL QUESTION PAPER					
ELECTROMAGNETIC WAVES AND TRANSMISSION LINES					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Phase and Group Velocities.	1	1	2
	b).	State the conditions for Lossless lines and distortion less lines.	1	1	2
	c).	Compute the reflection coefficient of a 50 Ω line terminated with 50+j75Ω impedance.	1	3	2
	d).	Discuss the input impedance of a matched line.	1	2	2
	e).	State the applications of Gauss's law.	2	1	2
	f).	Define equipotential surfaces?	2	1	2
	g).	Discuss the scalar magnetic potential.	3	2	2
	h).	State Faraday's law of electromagnetic induction.	4	1	2
	i).	Define a uniform plane wave and mention its characteristics.	5	2	2
	j).	Define polarization and list different types.	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Deduce the transmission line equations in terms of V and I.	1	3	5
	b).	Given R= 10.4 Ω/m, L =0.00367 H/m, G = 0.8 x 10 ⁻⁴ mhos/m,C = 0.00835 μF/m. Calculate characteristic impedance and propagation constant at 1.0 KHz.	1	3	5
		OR			
3.	a).	Deduce the expression for characteristic impedance of a line in terms of primary constants.	1	3	5
	b).	Explain how a finite length transmission line terminated by its characteristic impedance acts as an infinite line.	1	3	5
		UNIT-2			
4.	a).	Deduce the expression for input impedance in terms of secondary constants.	1	3	5
	b).	Discuss about reactance properties of open and short circuits lines of	1	2	5

		different lengths.			
		OR			
5.	a).	Discuss in detail about impedance matching with Quarter wave transformer.	1	2	5
	b).	Explain the construction of Smith chart and mention its applications.	1	2	5
		UNIT-3			
6.	a).	State Gauss's law. Deduce the expression for Electric field due to an infinite line charge.	2	3	5
	b).	A charge $Q = -10\text{nc}$ is at the origin in free space. If the x-component of $\mathbf{E}$ is to be zero at the point (3,1,1), determine the value Q_t which should be kept at the point (2,0,0).	2	3	5
		OR			
7.	a).	Deduce the expression for electric field due to an electric dipole.	2	3	5
	b).	Define Electric potential. Given the potential field $V = 2xy^2 - 5z$. Find the potential, electric field intensity, electric flux density and volume charge density at P(-4,3,6).	2	3	5
		UNIT-4			
8.	a).	Explain about Biot Savart's law in detail.	3	2	5
	b).	Deduce the boundary conditions for $\mathbf{E}$, $\mathbf{H}$, $\mathbf{D}$ and $\mathbf{B}$ at the interface between two different dielectric media.	4	3	5
		OR			
9.	a).	Deduce the expressions for Maxwell's equations in both differential and integral form.	4	3	5
	b).	The Magnetic field intensity in a region of space is $\mathbf{H} = (x+2y)/z^2 \mathbf{a}_y + 2/z \mathbf{a}_z$ A/m. Determine $\mathbf{J}$ and also the total current passing through the surface $Z=4$, $1 \leq X \leq 2$, $3 \leq Y \leq 5$.	3	3	5
		UNIT-5			
10.	a).	Explain the wave propagation in good dielectrics.	5	2	5
	b).	State wave propagation characteristics of good conductors. A medium is defined by $\sigma = 5.8 \times 10^7 \text{ mho/m}$, $\epsilon_r = 1$, $\mu_r = 1$, and supports a uniform plane wave of frequency 60Hz. Determine the attenuation constant, propagation constant, intrinsic impedance, wavelength and phase velocity.	5	3	5
		OR			
11.	a).	Explain about linear and circular polarizations of a uniform plane wave.	5	2	5
	b).	State and prove Poynting theorem.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC2203					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. II Semester MODEL QUESTION PAPER					
ELECTRONIC CIRCUIT ANALYSIS					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Discuss the limitations of h-parameters.	1	2	2
	b).	Define coupling. list out the types of coupling.	2	1	2
	c).	Discuss the advantages of multistage amplifiers.	2	2	2
	d).	List out the various types of feedback topologies.	3	1	2
	e).	Discuss the merits and demerits of negative feedback amplifiers.	3	2	2
	f).	State Barkhausen criterion for positive feedback amplifiers.	4	1	2
	g).	State the advantages of crystal oscillators.	4	1	2
	h).	State the conversion efficiencies of Class-A, B & AB Power Amplifiers.	5	1	2
	i).	Classify the tuned amplifiers.	5	2	2
	j).	Explain the significance of the Q-factor in tuned amplifiers.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Determine high frequency parameters in terms of low-frequency parameters.	1	2	5
	b)	Obtain short circuit current gain for CE amplifier at high frequency.	1	3	5
		OR			
3.		Analyze the common source amplifier at high frequencies,	1	4	10
		UNIT-2			
4.	a).	Derive the expressions for voltage gain of an RC coupled amplifier using BJT at low frequency.	2	3	5
	b).	A transistor is connected as a CE amplifier with load resistance of 10K Ω . The parameters are h_{ie} =5K Ω and h_{fe} =330. Calculate the overall gain for mid frequency range when four such stages are connected in cascaded RC coupling. Assume R_s =0.	2	3	5
		OR			
5.	a).	A two stage RC coupled amplifier uses the transistor whose h-	2	3	5

		parameters are $h_{ie}=4.5\text{ K}\Omega$, $h_{fe}=330$. If the load resistance $R_L=5.5\text{ K}\Omega$ then find the required value of coupling capacitor C_C , so that the lower cut-off frequency is 60 Hz.			
	b).	Discuss the various methods of coupling in multistage amplifiers.	2	2	5
		UNIT-3			
6.		Analyze voltage series feedback amplifier.	3	4	10
		OR			
7.	a).	Derive the effect of negative feedback on gain.	3	3	5
	b).	A negative feedback of $\beta=0.002$ is applied to an amplifier of gain 1000. Calculate the change in overall gain of the feedback amplifier, if the internal amplifier is subject to a gain reduction of 15%.	3	3	5
		UNIT-4			
6.	a).	Explain the working of Hartley oscillator using BJT and derive frequency of oscillations.	4	3	5
	b).	In a colpitt's oscillator $C_1=0.02\text{ }\mu\text{F}$ and $C_2=0.2\text{ }\mu\text{F}$. if the frequency of the oscillator is 10KHz, Find the value of the inductor and also find the required gain for oscillator.	4	3	5
		OR			
9.		Obtain the frequency and condition for sustained oscillations of RC phase shift oscillator.	4	3	10
		UNIT-5			
10.		Explain the working of class B push-pull power amplifiers and calculate its efficiency. Discuss its merits and demerits	5	3	10
		OR			
11.		Obtain the frequency response of double tuned amplifiers	5	3	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EC2204					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. II Semester MODEL QUESTION PAPER					
ANALOG COMMUNICATIONS					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define modulation.	1	1	2
	b).	Explain frequency translation.	1	2	2
	c).	Compare the advantages and disadvantages of DSB-SC and SSB.	2	2	2
	d).	What is Quadrature Amplitude Modulation?	2	1	2
	e).	What is the effect of modulation index on bandwidth in FM.	3	1	2
	f).	What is pre-emphasis and de-emphasis? Why is it used?	3	1	2
	g).	Define selectivity, sensitivity.	4	1	2
	h).	What are the main functions of an RF amplifier in a Radio Receiver?	4	1	2
	i).	Define resistor noise and shot noise.	5	1	2
	j).	Define signal to noise ratio & figure of merit.	5	1	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Explain the need for modulation.	1	2	5
	b).	Explain the generation of AM wave using square law modulator with a neat circuit diagram.	1	3	5
		OR			
3.	a).	Explain the demodulation of AM signal using envelope detector.	1	3	5
	b).	Calculate the modulation index of a certain AM transmitter which radiates 10KW with carrier unmodulated, and 10.725 KW when the carrier is sinusoidally modulated. Also determine its total radiated sideband power.	1	3	5
		UNIT-2			
4.	a).	Explain the generation of DSB-SC wave using Balanced modulator.	2	3	5

	b).	Explain the coherent detection of DSB-SC modulated waves.	2	2	5
		OR			
5.	a).	Explain the generation of SSB signal using phase shift method.	2	3	5
	b).	Explain Vestigial Side Band Modulation and demodulation	2	2	5
		UNIT-3			
6.	a).	Derive and explain Frequency deviation and Phase deviation in Frequency Modulation. Give the relationship between FM and PM with the help of neat block diagrams.	3	3	5
	b).	Find the modulation index, maximum frequency deviation and bandwidth of FM wave represented as $v(t)=10 \cos (3180 t + 6\sin(1250\pi t)) \text{ volts}$ What power will be dissipated by this FM wave in a 20Ω resistor?	3	3	5
		OR			
7.	a).	Explain the generation of frequency modulation using Armstrong method with a neat diagram.	3	3	5
	b).	Explain the operation of a Phase Locked Loop FM demodulator with a neat diagram.	3	3	5
		UNIT-4			
8.	a).	Explain the operation of AM transmitter with the help of a neat block diagram.	4	2	5
	b).	Explain the operation of FM transmitter with the help of a neat block diagram.	4	2	5
		OR			
9.	a).	Explain the principle of operation of a super heterodyne radio receiver with the help of a neat block diagram.	4	2	5
	b).	Explain the principle of operation of a communication receiver with the help of a neat block diagram.	4	2	5
		UNIT-5			
10.	a).	Derive the expression for signal -to-noise ratio in DSB-SC systems.	5	3	5
	b).	Explain the modulation and demodulation techniques for pulse time modulation systems.	5	2	5
		OR			
11.	a).	Derive an expression for figure of merit in FM systems.	5	3	5
	b).	Explain Time Division Multiplexing.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

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